



Tamim Chowdhury

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Summary

A Computer Science undergraduate student specializing in Agentic AI, Explainable Machine Learning, and Interactive Web/Mobile Systems. Experienced in architecting autonomous multi-agent platforms (LangGraph, Vertex AI, Google ADK) and building explainable ML systems (XGBoost, SHAP) for atmospheric forecasting and material informatics (RDKit). Passionate about developing immersive 3D interfaces and scalable, data-driven applications.

Skills

Programming languages:	Lan-	Python, C/C++, Java, Kotlin, TypeScript, SQL, Haskell
AI / ML:		LangGraph, Google ADK, Vertex AI, Scikit-learn, TensorFlow, PyTorch, XGBoost, SHAP, RDKit, Matminer, MediaPipe, Gemini API, Ollama, RAG Systems, Prompt Engineering, Agentic AI
Backend & APIs:		FastAPI, Node.js, REST API Development, JWT Authentication
Frontend:		React, Next.js, React Flow, Three.js, HTML/CSS
Databases:		PostgreSQL (PGVector), Redis, MongoDB, MySQL, SQLite
DevOps / Tools:		Git, Docker, Android Studio, GitHub Actions
Courses:		Introduction to Programming, Data Structures and Algorithms, Advanced Programming, Operating Systems, Machine Learning, Analysis and Design of Algorithms, Usable Security and Privacy, Mobile and Ubiquitous Computing, Maths I, Maths III, Maths IV, Probability and Statistics, Discrete Mathematics, Information Integration and Applications, Fundamentals of Database Systems, Computer Organisation, Basic Electronics, Digital Circuits, Practical Bioinformatics, Computing for Medicine, Haskell Mock, Introduction to HCI

Projects

B.Tech Thesis Project (BTP): Delhi Pollution Trapping Early Warning System <i>Advisor: Dr. Anuj Grover</i>	Individual (Jan, 26 - Present)
- Engineered an explainable ML system to forecast 6-hour-ahead surface atmospheric pollution trapping events (stagnations where $\text{PM}_{2.5} > 250 \mu\text{g}/\text{m}^3$ and Ventilation Coefficient $< 1500 \text{ m}^2/\text{s}$), achieving an 87.4% F1-score (XGBoost). - Integrated multi-source real-world data: 5 years of CPCB hourly air quality records, NASA FIRMS VIIRS satellite thermal hotspots (Fire Radiative Power), IGI Airport METAR visibility/fog records, and road traffic congestion models. - Implemented a meteorological deweathering framework (using XGBoost/HGBDT) achieving $R^2 = 0.87$ to separate emission-driven pollution trends from weather-driven stagnation, utilizing SHAP explainability for physical validation. - Technologies: Python, Scikit-learn, XGBoost, HGBDT, SHAP, Pandas	

Aura RAG Enterprise & AgentFlow AI

Advanced RAG & Multi-Agent Platforms

Individual
(Mar, 26 - Present)

- **Aura RAG:** Built an enterprise-grade document Q&A system using **PostgreSQL/PGVector**, Redis caching, and FastAPI.
- **AgentFlow AI:** Engineered an autonomous research platform using **LangGraph**, FastAPI, and React v19, coordinating 6 specialized AI agents with parallel search pipelines and Human-in-the-Loop gates.
- **Technologies:** FastAPI, LangGraph, React, PostgreSQL (PGVector), Redis, Node.js

NutriMind OS (Predictive Nutrition AI)

Vertex AI Agent Engine

Team Size-1
(May, 26)

- Engineered a multi-agent behavioral nutrition OS using **Google ADK** and **Gemini 3 Pro/Flash**.
- Integrated Google Fit via Managed MCP and implemented Vertex AI Code Execution Sandbox for dynamic diet computations.
- **Technologies:** Google ADK, Vertex AI, Gemini API, Android Studio

AI-Based Prediction of Material Properties

Team Size-2
(Nov, 25 - Present)

- Built ML models to predict mechanical, thermal, and adsorption properties of polymers and composites.
- Used Materials Project, AFLOW, Polymer Genome for data; extracted descriptors using RDKit and Matminer.
- Trained Random Forest, Gradient Boosting, Neural Networks; validated with experimental datasets.
- **Technologies:** Python, Scikit-learn, TensorFlow, Matminer, RDKit

Zuri AI Chatbot & SmartGuard Android App

Advisor: Arun Balaji Buduru

Individual
(Dec, 25)

- **Zuri AI:** Built a private-first AI chatbot application toggling between Google Gemini (Cloud) and Ollama (Local).
- **SmartGuard:** Developed a robust Android security application using intelligent motion detection and automated background protocols to capture evidence.
- **Technologies:** Java, Kotlin, Android Studio, SQLite, Gemini API, Ollama

Smart Water Bottle App

Advisor: Pragma Kar

Team Size-2
(Feb - Apr, 26)

- Designed and developed an Android application to track hydration and sync with a smart water bottle hardware via Bluetooth.
- Implemented real-time tracking, push notifications, and historical data logging using local databases.
- **Technologies:** Java, Android Studio, SQLite, Bluetooth API

Face Recognition REST API & Systems Programming

Systems & API Development

Individual
(Dec, 25)

- **Face Recognition:** Deployed a JWT-secured FastAPI backend using MediaPipe and SQLite, supporting batch image processing.
- **Systems:** Built a custom C-based ELF loader to map segments into memory, and a priority-based task scheduler.
- **Technologies:** FastAPI, MediaPipe, C, SQLite

SafePass Vault & Service Booking Apps

Advisor: Arun Balaji Buduru & Mukesh Mohania

Team Size-4,2
(Aug - Nov, 25)

- **SafePass:** Deployed a local-first password vault Android application using Kotlin, securing local credential storage with encrypted SQLite databases.
- **Service Booking:** Engineered a GUI-based multi-user booking program in Python, synchronized across multiple devices utilizing a centralized MySQL database.
- **Technologies:** Kotlin, Python, SQLite, MySQL, Android Studio

Education

Indraprastha Institute of Information Technology, Delhi
B.Tech (CSE)
2023 - 2027

CGPA: 7.15
(Until Last Semester)

Notre Dame College, Dhaka
HSC
2020

GPA: 5.00

R. B. Govt. High School, Joypurhat
SSC
2018

GPA: 5.00

Certifications

- **Ethical Hacking**
- **Haskell – Mock Course**
- **AI Workflow: Enterprise Model Deployment**

Research Interests

- **Agentic AI & LLM Orchestration:** Autonomous multi-agent architectures, Retrieval-Augmented Generation (RAG), and human-in-the-loop AI workflows.
- **Explainable Machine Learning & Scientific ML:** Interpretable ML models (SHAP) applied to atmospheric dynamics, environmental forecasting, and polymer/chemical property prediction (RDKit).
- **Interactive Systems & HCI:** Immersive 3D web environments, physics-based simulations, and IoT-integrated mobile applications.